HARI KIRAN VENTRAPRAGADA

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PROFESSIONAL SUMMARY

Software Engineer with hands-on production experience across backend, full-stack, and applied AI/ML systems, built over three internships at an AI startup. Shipped a real-time WebSocket presence system merged to production and authored 10+ root-cause-analysis reports across chat, audio/video, and project-management modules. Comfortable across the full SDLC with Python, FastAPI, REST APIs, and React, with additional applied experience in RAG pipelines, knowledge graphs, and multi-agent LLM orchestration (LangGraph).

EXPERIENCE

Software Development Intern — OmniqAI Pvt Ltd, Bengaluru *Dec 2025 – Jul 2026*

- Designed and implemented a real-time user presence system (online/away/offline/in-call states) for a production chat platform using WebSockets; merged to production.
- Performed root-cause analysis on production issues across chat, audio/video, and project-management modules — including duplicate-message rendering, AI-response rendering, and in-call presence desync — authoring 10+ RCA documents with fixes implemented by the engineering team.
- Built the React frontend and backend for a Secrets & Keys management feature for secure storage of API keys and credentials.

AI R&D Intern — OmniqAI Pvt Ltd, Bengaluru *May 2025 – Jun 2025*

- Contributed to an MCP (Model Context Protocol) Agent Orchestration proof-of-concept integrated with Mailcow, using FastAPI and Docker, with Playwright for automated testing and SonarQube for code-quality review.

AI Research Intern — OmniqAI Pvt Ltd, Bengaluru *Dec 2024 – Jan 2025*

- Built a Knowledge Graph microservice (Neo4j, Python, Ollama) to construct knowledge graphs from Slack conversational data as a grounding layer for a Retrieval-Augmented Generation (RAG) system, applying prompt-engineering techniques (chain-of-thought, one-shot, few-shot).
- Containerized and deployed the service using Docker Swarm and Gitea CI; monitored service health with Prometheus.

PROJECTS

Autonomous Multi-Step AI Agent | github.com/Harihkvent/Autonomous-Multi-Step-AI-Agent
Python, FastAPI, LangGraph, React/Vite

- Built a full-stack agent platform decomposing natural-language tasks into multi-step plans, executed via a tool registry (web search, calendar, email, document generation), with a sequential Planner → Executor → Validator → Retry pipeline and a multi-agent LangGraph supervisor.
- Later self-reviewed the codebase and fixed five production-grade bugs, including a cross-user data leak via global state, unsafe concurrent disk writes, and an unsafe eval()-based calculator replaced with AST sandboxing.

CareerPath AI — Career Advisory & Interview Platform
LangGraph, FastAPI, Redis, MongoDB, React, AWS

- Built a multi-agent career-advisory platform matching resumes against 63,000+ job roles with ML-based scoring, skills-gap analysis, and AI-generated, milestone-based learning roadmaps.
- Implemented a multi-round AI mock-interview system with resume-tailored questions and real-time LLM evaluation, backed by JWT auth and a MongoDB/Beanie data layer; used Redis to coordinate agent state; deployed on AWS (EC2, S3, RDS).

AI Calorie Tracker | caltrack-bay.vercel.app
Expo/React Native, FastAPI, Supabase, Gemini API

- Built and deployed a cross-platform nutrition-tracking app using the Gemini API to identify food and estimate calorie/macro content from photos and text, validating AI-generated outputs against an expected schema before persisting them.

PUBLICATION

- Co-authored a peer-reviewed paper (IJSREM) on a multi-agent AI career-advisory platform combining resume analysis, hybrid job matching, and mock-interview evaluation; the hybrid retrieval matcher achieved Precision@5 of 0.79 across 100 resume-job pairs.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL, C++, C
- **Backend & APIs:** FastAPI, Flask, Node.js/Express, REST APIs, Microservices, WebSockets, JWT Authentication
- **AI/ML & GenAI:** RAG, LangChain, LangGraph, Multi-Agent Orchestration, Prompt Engineering, Semantic Search, MCP (Model Context Protocol)
- **Databases:** SQL, MySQL, MongoDB, Neo4j, Redis
- **Cloud & DevOps:** Docker, Docker Swarm, AWS (EC2, S3, RDS, Lambda), Git/GitHub, CI/CD, Linux
- **Frontend:** React, React Native, TypeScript, TailwindCSS
- **Core CS Foundations:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

B.Tech, Information Technology — MVGR College of Engineering, Autonomous *2022 – 2026*
CGPA: 7.76 / 10